

2025

(Session : 2024-27)

(Paper ID : 12007)

Time : 3 hours

Full Marks : 80

Candidates are required to give their answers in their own words as far as practicable.

The questions are of equal value.

Answer any five questions.

1. Solve the following:

(a) Convert the following decimal numbers to binary, octal and hexadecimal:

(i) 156

(ii) 1023

(b) Convert the following binary numbers to decimal, octal and hexadecimal:

(i) 11010101

(ii) 1010111111

2. (a) Perform binary addition and subtraction:

(i) $1011 + 1101$

(ii) $10001 - 1010$

(b) (i) Convert the hexadecimal number 1A3F to binary and decimal.

(ii) Convert the binary number 111010.101 to decimal and hexadecimal.

3. Explain the Von Neumann Architecture with a neat diagram.

4. Discuss pipelining in processors. What are pipeline hazards and how can they be reduced?

5. Explain memory hierarchy in detail, including cache, main memory and secondary memory.

6. Explain the concept of virtual memory. How is paging used to implement it?

7. Discuss various input / output techniques, including programmed I/O, interrupt-driven I/O and DMA.

8. What is Boolean Algebra? Explain its importance in digital logic design.

9. What is a register in digital electronics? Explain its purpose.

10. Define different digital or combinational logic circuit.